



2026 Tomorrow Trials – Competition Pack

Thank you for entering our first ever Tomorrow Trials competition. We hope that you get a lot out of this experience, and most importantly, we hope that you have fun and enjoy working on something a bit different.

This document aims to provide the information you will need to start working on your competition entries. If you have any questions that are not answered in this pack, please speak with HR.

The Brief:

Each team is tasked with designing, building, and programming a swarm of four autonomous drones that will race in a fixed box formation through a defined obstacle course. The drones must autonomously navigate from Point A to Point B, maintaining swarm integrity throughout the course using AI developed by the team.

The challenge combines hardware design, control systems, AI, and teamwork, and is intended to reward both speed and precision of swarm behaviour.

Competition Objective:

Goal: To complete the obstacle course in the fastest possible time while maintaining the most accurate swarm formation.

Competition Deadline:

The deadline for submitting your entries is 20th November 2026. Further details will be provided to confirm the date, time and location for the actual race.

Team Composition:

Teams have been created to ensure a fair split by grade, capability and experience. It is up to each team to assign tasks and responsibilities as they see fit.

Drone Specifications:

- Each team must submit exactly 4 drones.
- Each drone must fit within a maximum bounding box of 250 mm × 250 mm × 250 mm (including propellers, propeller guards, antennae, landing gear/feet, etc).
- Maximum take-off weight per drone: 1.5 kg.
- Drone frames may be purchased pre-made or custom-built.
- Motors, ESCs, batteries, and flight controllers may be commercially sourced.



- Any repairs or replacements during the competition must remain within the original specifications.
- Propeller guards are mandatory (and must comply with the above dimensions).
- All drones must include a clearly accessible physical kill switch or equivalent failsafe.
- Battery type must be LiPo only, with a maximum of 4S (4 x 3.7V cells nominal).

Autonomy & AI Requirements:

- Full autonomy is mandatory. No manual or remote piloting is allowed at any stage of the run. (The drones must start and end the course autonomously, with no manual intervention during the run).
- Navigation, obstacle avoidance, and swarm coordination must be controlled by AI and control logic developed by the team.
- Use of open-source libraries and frameworks is permitted.
- Pre-trained models are allowed only if significantly adapted and integrated by the team.
- Decision-making, navigation logic, and swarm behaviours must be demonstrably authored by team members.

Swarm & Formation Rules:

- Drones must maintain a 3D box formation throughout the course.
- The dimensions of the 'box' can be set by the team, and can change in size as required to best meet the course, but a box structure should try to be maintained throughout.
- Temporary deviations from the formation are allowed only when avoiding obstacles, but the formation must be re-established within 2 seconds.

Sensing, Communication & Compute Constraints:

- Each drone must carry its own onboard computer capable of running navigation, control, and AI logic in real time.
- Inter-drone communication is permitted using Wi-Fi, radio, optical, blue tooth or acoustic methods.
- A single "leader" drone architecture is allowed, but hard dependency on a single external ground station is not.
- No external motion capture systems (e.g. Vicon, OptiTrack) or third-party tracking infrastructure may be used during a run.
- No communications external to the drones is allowed during the run.



- GPS may be used if available indoors, but teams should assume unreliable or degraded GPS signals.

Obstacle Course Specification:

- The course will be an indoor, bounded environment.
- A Track will be laid out on the floor using clearly identifiable tape (such as black/yellow striped tape or similar), in addition a number of obstacles will be included.
- Obstacles may include:
 - Vertical gates
 - Horizontal bars
 - Narrowing “corridors”
 - Tunnels
 - Elevation changes
 - Partial occlusions
- Minimum clearance through any opening: 200 mm per drone, but formation clearance may be tighter.
- Course dimensions will be published at least 4 weeks before execution.
- The start box and end box will be clearly defined and sized to accommodate the full swarm formation.

Run Conditions:

- Each team will be given two official runs.
- The best valid run will count toward final scoring.
- A run starts when the first drone leaves the start box.
- A run ends when all four drones enter the finish box and come to a stop.
- Maximum run time: 5 minutes. Exceeding this counts as a DNF.

Available Resources:

Each team will have a budget of £10,000 to manage.

- All purchases must directly relate to the competition, receipts will be required by the finance team.
- Teams are permitted an additional £200 per person for refreshments for team meetings.



- Each team member can book up to 37.5 hours to the preparation, the booking code '2026 Tomorrow Trials' should be used (within HR).
- Teams are able to use tools from the production department but must get permission from the Workshop Manager in advance and must be supervised when using them.
- Existing engineering software currently held by Stirling can be used, with license allocation being managed in the normal way (requested through engineering and IT, and allocated on a priority basis).

Safety Rules:

- All drones must pass a pre-flight safety inspection, assessed by the judging team.
- Any drone exhibiting unstable, unsafe, or uncontrolled behaviour will trigger an immediate run abort.
- Emergency stop authority rests with competition officials at all times.
- No sharp edges, or damaged propellers are allowed.
- All wiring to be secured safely and to have appropriate electrical insulation.
- Batteries must be secured and fire-safe at all times.
- Teams must follow all local indoor flight safety rules.

Team Presentation:

Each team must also give a presentation.

- All team members must deliver part of the presentation.
- Slides are optional but recommended. Creativity is encouraged.
- The presentation must be 12 minutes long, plus 3 minutes for Q&A.
- Each presentation must cover the following:
 - Brief summary of your solution.
 - How you worked together as a team – did you assign roles or jointly work on each task? How did you make decisions?
 - How you managed the budget?
 - How did you manage your time?
 - Did you encounter any problems, if so, how did you overcome these?
 - What did you learn from this project? Would you approach it differently if you were to do it again?



Scoring:

75% of the scoring will be derived from the race, with the remaining 25% from the presentation. Please see a full breakdown of the scoring below.

Criteria	Points
Completion of race	
All 4 drones complete the course autonomously	1
No manual intervention or override	1
Formation maintained within rules for majority of the run	1
<i>Maximum available points</i>	<i>3</i>
Course completion time for all 4 drones	
Fastest run	6
2 nd fastest	3
3 rd fastest	1
Failure to complete	0
<i>Maximum available points</i>	<i>6</i>
Swarm formation accuracy	
Formation consistently tight and stable	6
Minor deviations, rapid recovery	4
Frequent deviations, recovery usually adequate	3
Formation maintained intermittently	2
Formation rarely coherent	1
Formation fundamentally broken	0
<i>Maximum available points</i>	<i>6</i>
Obstacle strike	-2
Any contact, including with another drone	-2
Exceedance of drone weight allowance	-1
Presentation	
Teamwork and collaboration	1
Problem solving	1
Budget and resource management	1
Reflection and learning	1
Presentation delivery	1
<i>Maximum available points</i>	<i>5</i>

Good luck!!